CADETS · DAY 2 · LESSON 2 · DATA SCIENCE

Logistic regression and the odds ratio

From a yes/no outcome to a risk score, and how to read every coefficient as an OR

Dr. Rotem Gershon · DS2 · ~55 min lecture + ~25 min notebook

ISLP Ch. 4 (4.1–4.3)

WHERE WE ARE

Today’s lesson goals

By the end of this lesson you can

- Explain why a yes/no outcome needs logistic rather than linear regression
- Turn a coefficient into an odds ratio, for a binary and for a continuous predictor
- Tell a crude OR from an adjusted OR
- Say in one sentence what a loss function is
- Use a fitted model to give each patient a predicted probability

THE CLINICAL QUESTION

Which admitted patients are at high risk of dying in hospital?

2,000

internal-medicine admissions in the CADets dataset (release 1)

181

died in hospital: 9.1%. The outcome is yes/no (0/1)

THREE WAYS TO SAY THE SAME RISK

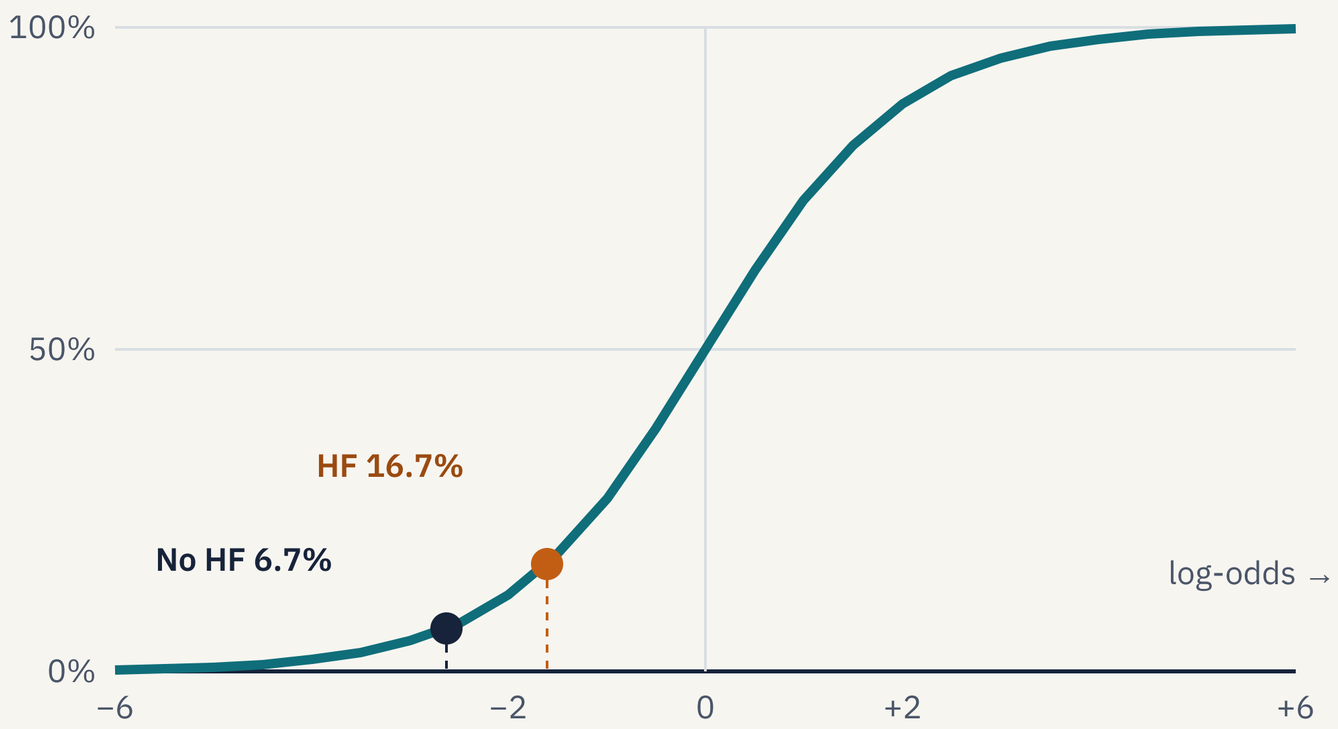
Probability, odds and log-odds

Example	Probability p	Odds = p / (1 – p)	Log-odds = ln(odds)
No heart failure: 103 of 1,533 died	6.7%	0.072	−2.63
Heart failure: 78 of 467 died	16.7%	0.20	−1.61
A coin toss	50%	1	0
Very high risk	90%	9	+2.20
Possible range	**0 to 1**	**0 to ∞**	**–∞ to +∞**

Log-odds have no floor and no ceiling, so a straight line can live on that scale without ever predicting an impossible risk.

THE LOGISTIC (SIGMOID) FUNCTION · ISLP 4.3.1

The S-curve turns any log-odds back into a risk




```
p = 1 / (1 + e^(−log-odds))
```

Log-odds 0 → risk 50%.

Very negative → risk near 0%. Very positive → risk near 100%.

The model does its arithmetic on the log-odds scale, then this curve converts the answer into a probability.

THE LINEAR STRUCTURE

The same recipe as DS1, on the log-odds scale

$$\text{log-odds(death)} = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \ldots + \beta_p \cdot x_p$$

	DS1 · Linear regression	DS2 · Logistic regression
Outcome	a number (length of stay)	yes/no (in-hospital death)
Left side	the outcome itself	the log-odds of the outcome
β for 1 unit more of x	adds β days	adds β to the log-odds
Readable as	β itself	**e^β = odds ratio**

A loss function scores how wrong the predictions are

What happened	Model’s predicted risk	Penalty (log loss)
Died	80%	0.22 · small
Died	10%	**2.30 · large**
Survived	10%	0.11 · small
Survived	80%	**1.61 · large**

Penalty = −ln(the probability the model gave to what actually happened).

Confident and wrong costs the most.

Fitting = choosing the β that make the observed outcomes most likely: maximizing the **log-likelihood**, which is the same as minimizing the **log loss** (cross-entropy). The same loss trains tree ensembles and neural networks.

PART 1 OF 3 · UNIVARIABLE

One binary predictor: heart failure

From the 2×2 table you already know to a model, and from the coefficient to an odds ratio.

RECAP FROM EPI1

The 2×2 table gives an odds ratio of 2.78

	Died	Survived	Odds
Heart failure	78	389	0.201
No heart failure	103	1,430	0.072

Odds ratio

`0.201 / 0.072`

= 2.78

Odds = died / survived in each row. OR = odds with heart failure ÷ odds without.

UNIVARIABLE MODEL · BINARY PREDICTOR

The model gives back the same odds ratio

```
log-odds(death) = −2.631 + 1.024 × heart_failure
```

Term	β	e^β	What e^β means
Intercept (β_0)	−2.631	0.072	Odds of death when heart_failure = 0
heart_failure (β_1)	1.024	**2.78**	**Odds ratio**: HF vs no HF

e^1.024 = 2.78, exactly the OR from the 2×2 table. With one binary predictor, the model and the table agree.

How to say it, and how not to

> “Patients admitted with heart failure had **2.8 times the odds** of dying in hospital compared with patients without heart failure (OR 2.78, 95% CI 2.03–3.81).”

Reading any OR

- OR > 1: higher odds · OR = 1: no difference · OR < 1: lower odds
- OR 2.78 = odds 178% higher
- Back to risk: 6.7% without HF, 16.7% with HF

Common mistakes

- “2.8 times the *risk*”: the risk ratio is 16.7 / 6.7 = 2.5
- “Heart failure *causes* death”: this is an association
- Reading β (1.024) as the effect: exponentiate first

PART 2 OF 3 · UNIVARIABLE

One continuous predictor: age

There is no 2×2 table for age. The model still gives an odds ratio, this time per unit.

UNIVARIABLE MODEL · CONTINUOUS PREDICTOR

Age: an odds ratio per year

```
log-odds(death) = −6.280 + 0.054 × age
```

Per 1 year of age

OR 1.056

e^0.054. Each extra year multiplies the odds of death by 1.056: **5.6% higher odds**.

Per 10 years of age

OR 1.72

e^(10 × 0.054) = 1.056^10. Ten years older: **72% higher odds**.

“Each additional 10 years of age was associated with 1.7 times the odds of in-hospital death (OR 1.72, 95% CI 1.51–1.95).”

INTERPRETING THE OR · CONTINUOUS PREDICTOR

Same OR, different change in risk

Age 40 → 50

Odds × 1.72

1.6% → 2.7%

+1.1 percentage points

Age 70 → 80

Odds × 1.72

7.6% → 12.4%

+4.8 percentage points

The OR is a **relative** measure: it stays fixed. The **absolute** change in risk depends on where the patient starts. Clinicians and product owners usually need both.

CONTINUOUS PREDICTORS · TWO PRACTICAL RULES

Pick a clinical unit, and check the shape

Predictor (alone)	OR per 1 unit	OR per clinical unit	Read as
Age	1.056 / year	1.72 / 10 years	older, higher odds
SpO_2	0.75 / 1% higher	1.34 / 1% lower	flip the direction
Respiratory rate	1.22 / breath/min	2.73 / 5 breaths/min	faster, higher odds

Warning: one OR assumes a straight line on the log-odds scale

Sodium alone gives OR 0.91 per mmol/L, as if higher sodium were always safer. Yet death is 40% at ≤128, 3.6% at 139–141 and 14% above 145 mmol/L: a U-shape one OR cannot describe (DS7).

PART 3 OF 3 · MULTIVARIABLE · ISLP 4.3.4

Many predictors at once: crude vs adjusted OR

Same equation, more terms. Each OR is now read “holding the other predictors constant”.

WHY ADD PREDICTORS?

Patients with heart failure are also older

Mean age, heart failure

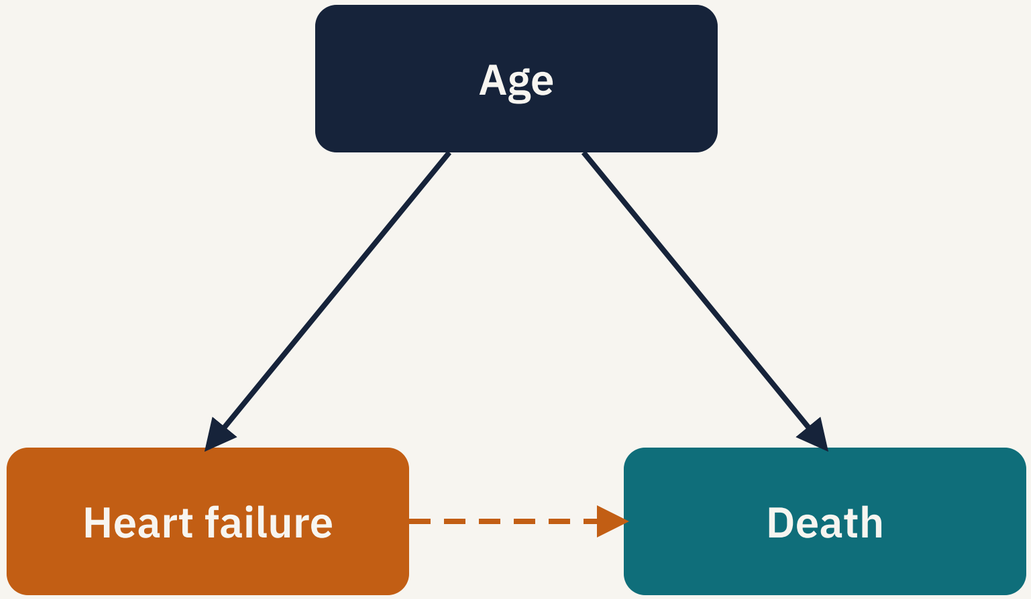
76

Mean age, no heart failure

68

Age raises the risk of death on its own. So the crude OR of 2.78 mixes two things: heart failure itself and the extra years that come with it.

Putting age in the model compares patients with and without heart failure **at the same age**.



MULTIVARIABLE MODEL · TWO PREDICTORS

Adjusted for age, the OR drops to 1.98

```
log-odds = −6.027 + 0.684 × heart_failure + 0.048 × age
```

Predictor	Crude OR	Adjusted OR	Adjusted for
Heart failure	2.78	**1.98**	age
Age (per year)	1.056	**1.049**	heart failure

> “**Among patients of the same age**, heart failure was associated with 2.0 times the odds of in-hospital death (adjusted OR 1.98, 95% CI 1.43–2.76).”

CRUDE VS ADJUSTED

The heart failure OR depends on what else is in the model

Crude

2.78

heart failure alone
95% CI 2.03–3.81

→

Adjusted for age

1.98

2 predictors
95% CI 1.43–2.76

→

Fully adjusted

1.47

14 predictors: age, sex, diseases, vitals, labs
95% CI 1.03–2.11

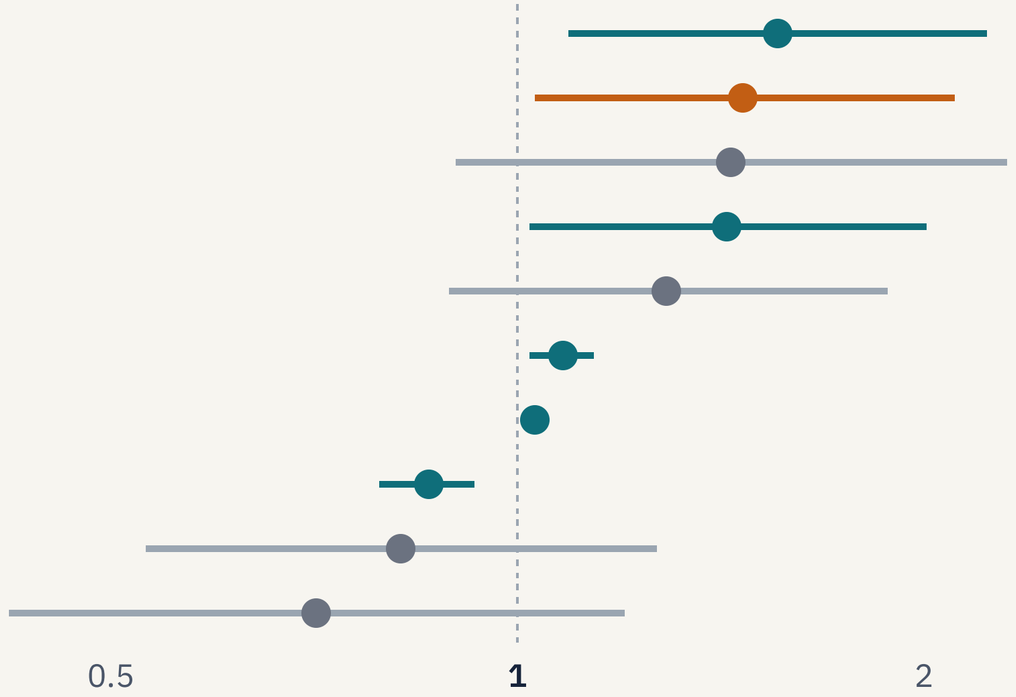
An adjusted OR is only meaningful together with its list of adjustments. **Always report what you adjusted for.**

THE FULL MODEL · ADJUSTED ORS WITH 95% CI

Reading a multivariable model at a glance

Creatinine (per 1 mg/dL)	1.56 (1.09–2.23)
Heart failure (yes vs no)	**1.47 (1.03–2.11)**
Active cancer (yes vs no)	1.44 (0.90–2.31)
Diabetes (yes vs no)	1.43 (1.02–2.01)
Male sex (vs female)	1.29 (0.89–1.88)
Resp. rate (per breath/min)	1.08 (1.02–1.14)
Age (per year)	1.03 (1.02–1.05)
SpO_2 (per 1% higher)	0.86 (0.79–0.93)
COPD (yes vs no)	0.82 (0.53–1.27)
CKD (yes vs no)	0.71 (0.42–1.20)

Grey = CI crosses 1 (no clear association)



One OR, two very different patients

Patient age	No heart failure	Heart failure	Odds ratio	Risk difference
60	4.1%	7.7%	1.98	+3.6 points
80	9.9%	17.8%	1.98	**+7.9 points**

The model has a single OR for heart failure, yet the added risk is twice as large in the 80-year-old, because that patient starts from a higher baseline.

For a decision about one patient, use the predicted risk, not the OR.

FROM MODEL TO PRODUCT · ISLP 4.3.3

Every patient gets a predicted risk

1 · The patient

80 years old, heart failure

2 · Add up the log-odds

$-6.027 + 0.684 \times 1 + 0.048 \times 80 = -1.53$

3 · S-curve

17.8%

- In code this is one line: **model.predict(df)** gives a risk for all 2,000 admissions.
- Sorting by predicted risk gives a worklist: “the 10 highest-risk patients today”.
- Whether the ranking is good enough to act on is the next lesson: DS3 (thresholds, AUROC, PPV).

PITFALLS

Four ways to misread a logistic model

OR is not a risk ratio

They are close only for rare outcomes. Heart failure: OR 2.8, risk ratio 2.5.

Adjusted is not causal

A prediction model ranks patients. Effects of treatments need causal design (CI1–CI3).

One OR assumes a straight line

On the log-odds scale. Sodium’s U-shape breaks it (DS7).

The unit hides the size

OR 1.03 per year of age is not small: it is 1.4 per decade. Always state the unit.

TAKEAWAYS

Five things to remember

1. Logistic regression is a linear model of the **log-odds**; the S-curve turns it into a risk between 0% and 100%.
2. **e^β is an odds ratio.** Binary predictor: yes vs no. Continuous predictor: per one unit, so choose the unit.
3. With one binary predictor, the model reproduces the 2×2 table OR exactly (2.78).
4. Adjusted ORs read “holding the others constant”; they change with the model (2.78 → 1.98 → 1.47).
5. The model is fitted by minimizing a **loss**, and its product is a **predicted risk** for each patient.

HANDS-ON · GOOGLE COLAB · 25 MIN

Notebook: DS2_core_logistic_regression.ipynb

Step 1 · 5 min

Heart failure: 2×2 table, then the same OR from a model

Step 2 · 5 min

Age: OR per year and per 10 years

Step 3 · 5 min

Heart failure + age: crude vs adjusted OR

Step 4 · 5 min

Full model: age, comorbidities, vitals and labs; table of ORs

Step 5 · 5 min

Predicted risk for every patient; the 10 highest-risk patients

You will write

One interpretation sentence for a binary and for a continuous predictor

AFTER CLASS

Reading and enrichment

ISLP, Chapter 4 (core)

4.1 An Overview of Classification
4.2 Why Not Linear Regression?
4.3.1–4.3.4 The Logistic Model, Estimating the Coefficients, Making Predictions, Multiple Logistic Regression

Enrichment notebook A

ISLP Lab 4.7 (logistic regression) on CADets data; loss functions in depth: squared error vs log loss, maximum likelihood (4.3.2)

Enrichment notebook B

Multinomial logistic regression for discharge destination (4.3.5); generalized linear models for length of stay (4.6)

Next in the data science track: **DS3 · Evaluation I**, which turns today's predicted risks into alerts, thresholds and an ROC curve.